



FASR Science use case: Quantifying CME-Driven Shock Properties Using Radio Imaging Spectroscopy

Shaheda Begum Shaik^{1,2}, Jason Kooi¹, Angelos Vourlidas³

¹*US Naval Research Laboratory, Washington, DC, USA*

²*George Mason University, Fairfax, VA, USA*

³*The Johns Hopkins University Applied Physics Laboratory, Laurel, MD, USA*

I. Science Goal(s)

This science case aims to quantify the physical properties and their evolution of CME-driven shocks in the solar corona using spatially resolved radio observations, along with complementary heliospheric and in situ measurements. Specifically, we seek to:

1. Localize CME-driven shock features close to the Sun during their initial formation and early evolution with FASR imaging,
2. Measure and constrain shock compression ratios and Alfvén Mach numbers using type II radio burst band splitting and combining with complementary lower-frequency radio observations when the split-band emission occurs below the FASR observing range, and derive the coronal magnetic field strength along the shock front.
3. Investigate the spatiotemporal evolution of these shock properties from their formation in the near-Sun coronal regions, and
4. Establish observational constraints linking shock evolution, as inferred from heliospheric imaging and in-situ measurements, to solar energetic particle (SEP) acceleration efficiency.

II. Scientific Rationale

(A) Scientific Importance

CME-driven shocks are the dominant drivers of large SEP events and play a central role in space weather. However, direct measurements of their plasma and magnetic field properties in the corona are sparse, and therefore critical shock parameters, such as Mach numbers and shock strength, remain poorly constrained.

White-light observations image shock sheaths and allow kinematic measurements, but rely on indirect methods to infer density compression [1][2], whereas in-situ measurements are limited to serendipitous single-point local sampling of the shock morphology beyond 10 solar radii. Radio type II bursts arise from plasma emission by shock-accelerated electrons in the corona (metric) and inner heliosphere (decimetric to kilometric). Some type II bursts exhibit band-splitting, which has been interpreted as emission from both upstream and downstream of the shock [3, 4, 5]. According to that interpretation, band splitting provides a direct measure of the shock compression ratio, given by $X = \frac{f_u^2}{f_l^2}$, where f_u and f_l are the upper and lower band fundamental frequencies. The compression ratio can then be used to derive the Alfvén Mach number through the Rankine–Hugoniot jump conditions. By combining these measurements with coronal density models (e.g., [6]), the magnetic field strength along the shock front can be estimated [7].

However, this interpretation is not universally accepted, and its potential as a shock diagnostic cannot be fully exploited [8, 9]. Current radio imaging capabilities lack spatial resolution, hampering the direct association of emission features within specific regions of the shock and preventing a conclusive interpretation for band splitting. This science case addresses this gap by using FASR imaging spectroscopy capabilities that uniquely enable spatially resolved measurements of shock properties, thereby allowing us to quantitatively link shock evolution to particle acceleration processes, including the efficiency of electron acceleration and the conditions for SEP production.

(B) Uniqueness to Capabilities

Current radio imaging spectrometers provide valuable CME and shock diagnostics, particularly at lower frequencies, but no existing facility offers the planned combination of broadband imaging spectroscopy needed to routinely localize early shock-related emission close to the Sun. FASR should be designed to uniquely enable this science through broadband, solar-dedicated imaging spectroscopy that provides simultaneous spatial and spectral resolution with high cadence and dynamic range. In combination with complementary lower-frequency instruments and white-light observations, FASR would help connect low-coronal shock formation and evolution with type II radio burst diagnostics, enabling spatially resolved studies of CME-driven shocks.

(C) Synergies

This science case will be significantly enhanced through synergy with radio diagnostics, along with complementary coronagraphic, heliospheric imaging, and in-situ observations. Current complementary facilities include OVRO-LWA, MWA, LOFAR, LASCO, STEREO, CCOR, Parker Solar Probe, and Solar Orbiter, which provide important observational foundations that will help guide and inform the next-generation instruments. Together, such datasets enable end-to-end characterization of CME-driven shocks from their formation in the corona to their propagation into the heliosphere.

FASR radio observations will primarily probe shocks in the low corona, where CME-driven shocks form and metric–decimetric type II radio emission is generated, while white-light and heliospheric imaging observations become increasingly important at larger heliocentric distances where shock-associated radio emission may become weak or absent.

(D) Measurements Required

Imaging measurements with frequency-dependent source localization are required to determine the spatial evolution and geometry of shock-associated radio emission in the low corona. These observations should primarily distinguish shock-related Type II plasma emission from other contemporaneous radio emissions associated with CME eruptions. High-cadence radio imaging spectroscopy is needed to track the propagation of shock features and relate the radio sources to CME morphology observed in complementary white-light observations. Polarimetric measurements, particularly circular polarization, would further help constrain the magnetic field and emission properties along the shock front.

Spectral measurements with sufficient bandwidth, spectral resolution, and temporal cadence are required to resolve type II radio burst lanes, identify possible band-splitting structures, and accurately track their frequency drift and temporal evolution. High signal-to-noise ratio and dynamic range are needed to detect faint split-band emission relative to the brighter background and downstream emission.

III. Science Requirements Tables

The observational requirements presented here are driven by the need to resolve and characterize CME-driven shocks during their formation and early evolution in the low (if possible, middle) corona. The proposed angular resolution (10–30 arcsec) is intended to distinguish spatial variations along the shock front and localize shock-associated radio emission relative to the CME structure. Sensitivity to large angular scales (5–30

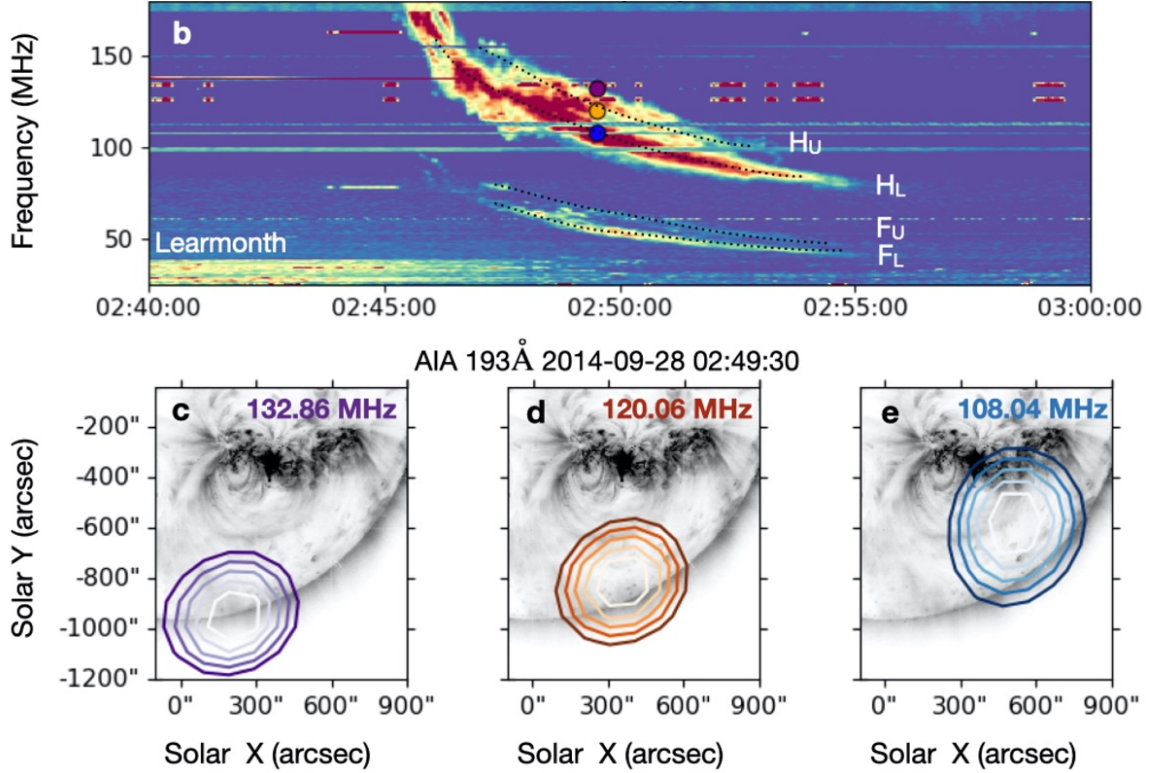


Figure 1: Figure adapted from [10]. (b) Dynamic spectrum of the type II radio burst observed by the Learmonth spectrograph, showing the split fundamental (FU, FL) and harmonic (HU, HL) bands. (c-e) MWA radio contours at 132.86 MHz (purple), 120.06 MHz (orange), and 108.54 MHz (blue), overlaid on an AIA EUV 193 Å base image. The contours (50–90% of the peak intensity) mark the locations of the radio sources, with the corresponding frequencies indicated by colored points in panel (b)

arcmin) and image sizes of 10–30 arcmin are required to capture the extended morphology of CME-driven shocks and their evolution.

Spectral resolutions of a few MHz and temporal cadences of a few seconds are motivated by the need to resolve type II burst lanes, potential band-splitting signatures, and their frequency drift. Type II band splitting is commonly characterized by relative frequency separations, for example $\delta f/f$ of 15% [10]. To accurately resolve the split bands, a spectral resolution of approximately 1–2% of the observing frequency (corresponding to a few MHz over the 0.2–1 GHz range) is desirable.

Dynamic range, sensitivity, and polarization requirements are driven by the need to detect weak shock-related emission and constrain magnetic field properties through spectropolarimetric diagnostics. Together, these specifications are intended to primarily enable quantitative measurements of shock compression, geometry, and magnetic field strength.

(A) OBSERVATIONAL TARGET		
Type of observation (what defines a ‘target’?)	imaging spectroscopy of Type II radio bursts	
Provide a brief description of the target	Type II radio bursts with band-splitting morphologies associated with CME events	
Number of targets	Primarily 1 - 2 high-speed ($\gg 1000 \text{ km s}^{-1}$) CME-driven shock type II events	lower coronal radio emission from shock front
Size of a single target (arcsec $\times$ arcsec)	Lower coronal regions	($\sim 10' \times 10'$ to $30' \times 30'$)
Distribution of all targets (arcmin $\times$ arcmin)	$\sim 10\text{--}30$ arcmin	~ 1.5 to $3 R_{\odot}$, if possible to $10 R_{\odot}$
Peak brightness (sfu/beam or Kelvin)	$\sim 10^6 - 10^9 \text{ K}$	
RMS brightness (sfu/beam or Kelvin)	$\sim 10^5 \text{ K}$	
Expected circularly polarized flux density	Stokes V (sfu/beam)	Detectable, but weak, $\gtrsim 1 \text{ sfu/beam}$
Expected circular polarization fraction	V/I	Required for magnetic field properties of shocks (Typically a few % e.g., [11])
Expected linearly polarized flux density	Stokes Q or U (sfu/beam)	low
Expected linear polarization fraction	Q/I or U/I	low

(B) SPECTRAL-TEMPORAL REQUIREMENTS	
Central Frequency (GHz)	0.2 - 1
Instantaneous Bandwidth (GHz/pol)	100 - 300 MHz
Spectral resolution (MHz)	a few MHz
Temporal resolution (seconds)	$\sim 15 - 30$

(C) POLARIZATION DATA PRODUCTS REQUIRED	
Stokes I required? (y/n)	yes
Stokes Q required? (y/n)	no
Stokes U required? (y/n)	no
Stokes V required? (y/n)	yes

References

- [1] Veronica Ontiveros and Angelos Vourlidas. Quantitative Measurements of Coronal Mass Ejection-Driven Shocks from LASCO Observations. , 693(1):267–275, March 2009. doi: 10.1088/0004-637X/693/1/267.
- [2] Ryun-Young Kwon and Angelos Vourlidas. The density compression ratio of shock fronts associated with coronal mass ejections. *Journal of Space Weather and Space Climate*, 8:A08, February 2018. doi: 10.1051/swsc/2017045.
- [3] B. Vršnak, H. Aurass, J. Magdalenic, and N. Gopalswamy. Band-splitting of coronal and interplanetary type II bursts. I. Basic properties. , 377:321–329, October 2001. doi: 10.1051/0004-6361:20011067.
- [4] B. Vršnak, J. Magdalenic, H. Aurass, and G. Mann. Band-splitting of coronal and interplanetary type II bursts. II. Coronal magnetic field and Alfvén velocity. , 396:673–682, December 2002. doi: 10.1051/0004-6361:20021413.
- [5] B. Vršnak, J. Magdalenic, and P. Zlobec. Band-splitting of coronal and interplanetary type II bursts. III.

(D) IMAGING REQUIREMENTS		
Required angular resolution (arcsec) (single value or range)	10–30 arcsec	
Largest angular scale required (arcsec)	~5–30 arcmin	
Mapped image size (arcmin $\times$ arcmin)	~10–30 arcmin	
Required pixel resolution (arcsec)	~5–10 arcsec	
Number of output/image channels	100–500	
Output bandwidth (min–max frequency, GHz)	0.3 – 3GHz - FASR-B	
Channel width (MHz)	1–5 MHz	
Required rms (sfu/beam or Kelvin) [per channel]	Sensitivity sufficient for faint up-stream emission	
Dynamic range within image	$\geq 100:1$	
Polarization accuracy (%)	a few %	
Zero spacing/total power required? (y/n)	n	
Required maximum latency (seconds, or N/A)	N/A	
Required flux density scale calibration accuracy	$\lesssim 10\%$	

Physical conditions in the upper corona and interplanetary space. , 413:753–763, January 2004. doi: 10.1051/0004-6361:20034060.

- [6] Yolande Leblanc, George A. Dulk, and Jean-Louis Bougeret. Tracing the Electron Density from the Corona to 1 au. , 183(1):165–180, November 1998. doi: 10.1023/A:1005049730506.
- [7] P. Kishore, R. Ramesh, K. Hariharan, C. Kathiravan, and N. Gopalswamy. Constraining the Solar Coronal Magnetic Field Strength using Split-band Type II Radio Burst Observations. , 832(1):59, November 2016. doi: 10.3847/0004-637X/832/1/59.
- [8] S. F. Smerd, K. V. Sheridan, and R. T. Stewart. On split-band structure in type ii radio bursts from the sun (presented by s.f. smerd). volume 57, page 389, 1974. URL <http://adsabs.harvard.edu/abs/1974IAUS...57..389S>.
- [9] R. A. Treumann and J. Labelle. Band splitting in solar type ii radio bursts. *The Astrophysical Journal Letters*, 399:L167–L170, November 1992. doi: 10.1086/186633.
- [10] Shilpi Bhunia, Eoin P. Carley, Divya Oberoi, and Peter T. Gallagher. Imaging-spectroscopy of a band-split type II solar radio burst with the Murchison Widefield Array. , 670:A169, February 2023. doi: 10.1051/0004-6361/202244456.
- [11] S. Normo, D. E. Morosan, P. Zhang, P. Zucca, and R. Vainio. Imaging and spectropolarimetric observations of a band-split type II solar radio burst with LOFAR. , 698:A175, June 2025. doi: 10.1051/0004-6361/202553702.